

Duality and KKT Conditions

Bounds, certificates, and inequality constraints

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**When a constraint may or may
not bind**

How do you certify “no better solution exists”?

An optimizer returns a feasible x with $f(x) = 10.003$. Is it optimal? Checking every other feasible point is impossible.

This lecture builds a second problem — the **dual** — whose value is a guaranteed lower bound on the best possible objective.

1. If the dual reports 10.000, no feasible point can beat 10.000.
2. The pair certifies: x is within 0.003 of optimal.

Support vector machines are trained through exactly this dual problem. The bound needs inequality constraints done right — we start there.

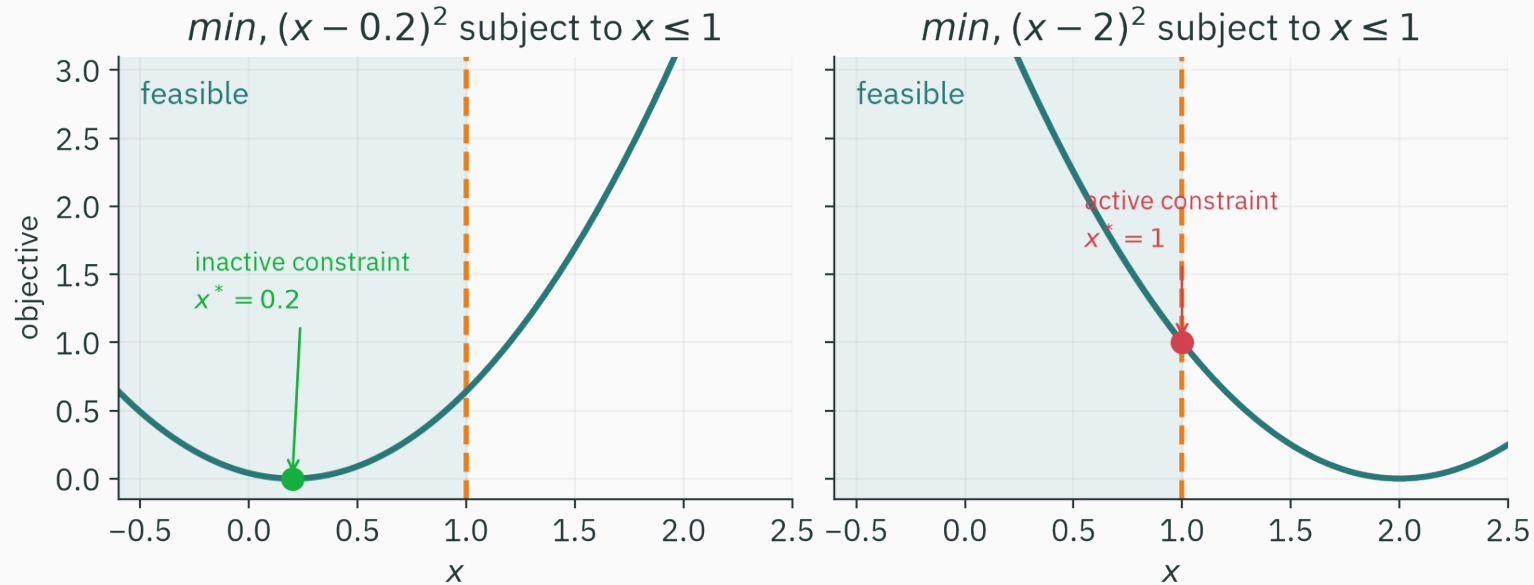
One boundary, two outcomes

Compare these problems:

$$\min_x (x - 0.2)^2 \quad \text{subject to} \quad x \leq 1,$$

$$\min_x (x - 2)^2 \quad \text{subject to} \quad x \leq 1.$$

The same inequality appears in both. In the first problem it does nothing; in the second it determines the answer.



1. **Inactive:** $x^* < 1$; there is positive slack.
2. **Active:** $x^* = 1$; the boundary is reached.

Put every inequality in one form

We will write a minimization problem as

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0.$$

For $x \leq 1$, use

$$g(x) = x - 1 \leq 0.$$

Its non-negative slack is

$$s = -g(x) = 1 - x \geq 0.$$

The multiplier must be non-negative

For $g(\boldsymbol{x}) \leq 0$, use

$$\mathcal{L}(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) + \lambda g(\boldsymbol{x}), \quad \lambda \geq 0.$$

At any feasible $\boldsymbol{x}$,

$$g(\boldsymbol{x}) \leq 0 \quad \Rightarrow \quad \lambda g(\boldsymbol{x}) \leq 0.$$

Therefore

$$\mathcal{L}(\boldsymbol{x}, \lambda) \leq f(\boldsymbol{x}).$$

This simple inequality is the source of the dual lower bound.

Prices for inequalities

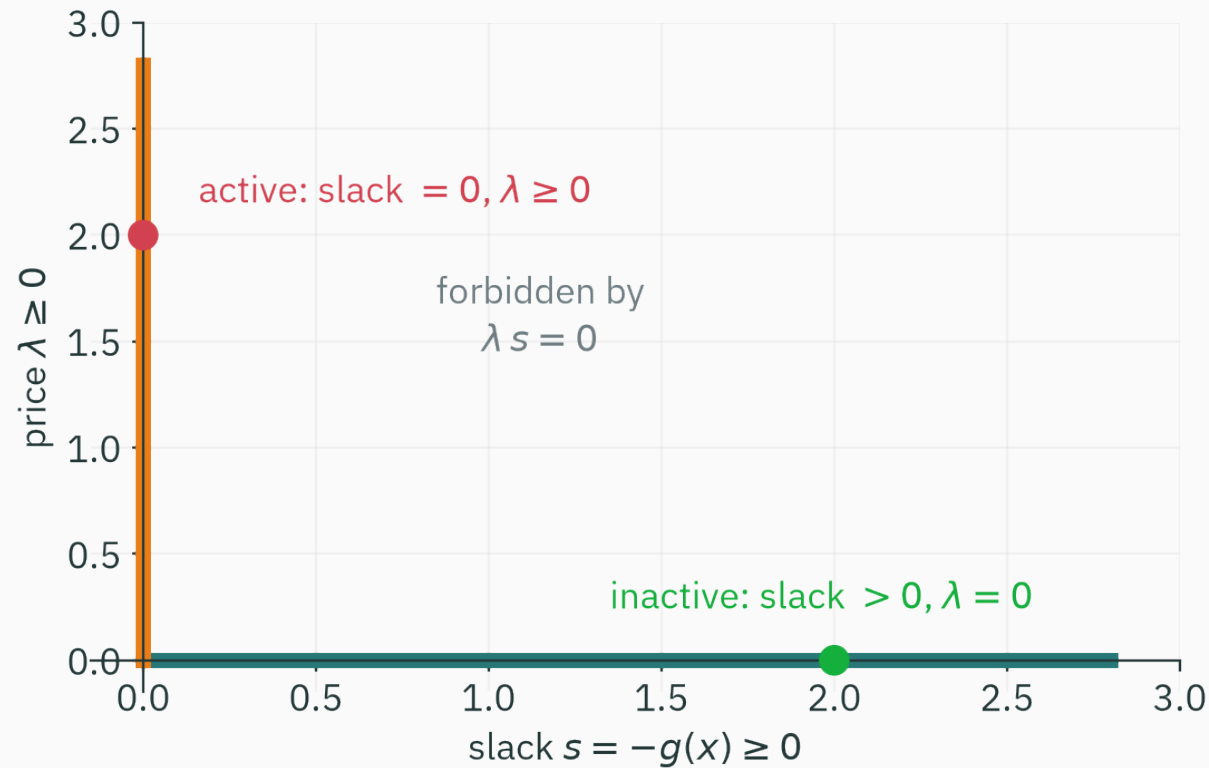
An active constraint can carry a positive multiplier: relaxing it can improve the optimum.

An inactive constraint has no local effect, so its multiplier should be zero.

These two cases are summarized by

$$\lambda \geq 0, \quad -g(\mathbf{x}) \geq 0, \quad \lambda g(\mathbf{x}) = 0.$$

The product equation is **complementary slackness**.



Either the constraint has slack, or it has a price. Their product must vanish.

Learning outcomes

By the end of this lecture you will be able to:

1. construct the Lagrange dual function,
2. prove ★ weak duality,
3. read a duality gap as an optimality certificate,
4. state and interpret the four KKT conditions,
5. verify KKT on an active and an inactive constraint,
6. and solve a small non-negative allocation problem from its KKT equations.

A lower bound from the Lagrangian

Start with the active example

$$p^* = \min_x (x - 2)^2 \quad \text{subject to} \quad x \leq 1.$$

We already know the constrained answer:

$$x^* = 1, \quad p^* = 1.$$

Write $g(x) = x - 1$ and

$$\mathcal{L}(x, \lambda) = (x - 2)^2 + \lambda(x - 1), \quad \lambda \geq 0.$$

Fix λ and minimize over x

For a chosen $\lambda \geq 0$, define the dual function

$$q(\lambda) = \inf_x \mathcal{L}(x, \lambda).$$

Differentiate with respect to x :

$$2(x - 2) + \lambda = 0 \quad \Rightarrow \quad x(\lambda) = 2 - \frac{\lambda}{2}.$$

Substitute back:

$$q(\lambda) = \lambda - \frac{\lambda^2}{4}.$$

Every $q(\lambda)$ is a lower bound

Take any feasible x . Since $\lambda \geq 0$ and $g(x) \leq 0$,

$$\mathcal{L}(x, \lambda) \leq f(x).$$

The infimum over all x cannot be larger than the Lagrangian at that feasible point:

$$q(\lambda) = \inf_z \mathcal{L}(z, \lambda) \leq \mathcal{L}(x, \lambda) \leq f(x).$$

In particular,

$$q(\lambda) \leq p^*.$$

For any primal-feasible x and dual-feasible $\lambda \geq 0$,

$$q(\lambda) \leq f(x).$$

Taking the best primal point on the right gives

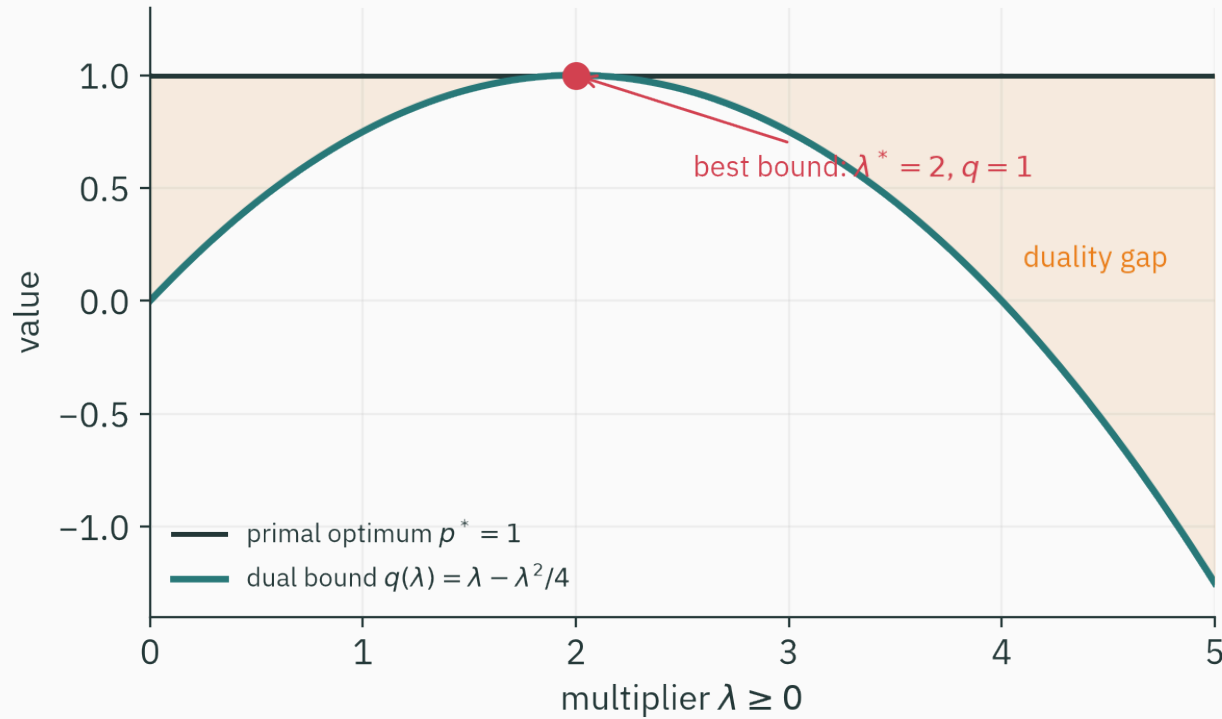
$$q(\lambda) \leq p^*.$$

Taking the best lower bound on the left gives

$$d^* = \max_{\lambda \geq 0} q(\lambda) \leq p^*.$$

This is weak duality. It holds even when the primal problem is non-convex.

Choose the best lower bound



Here $q(\lambda)$ reaches 1 at $\lambda = 2$. The dual lower bound meets the primal optimum.

Solve the dual directly

$$q(\lambda) = \lambda - \frac{\lambda^2}{4}, \quad \lambda \geq 0.$$

$$q'(\lambda) = 1 - \frac{\lambda}{2} = 0 \quad \Rightarrow \quad \lambda^* = 2.$$

$$q(\lambda^*) = 2 - \frac{2^2}{4} = 1 = p^*.$$

The associated minimizer of the Lagrangian is

$$x(\lambda^*) = 2 - \frac{\lambda^*}{2} = 1.$$

The duality gap

For a primal-feasible point x and a dual-feasible multiplier λ ,

$$\text{gap} = f(x) - q(\lambda) \geq 0.$$

Example: choose $x = 0.8$ and $\lambda = 1$.

$$f(0.8) = 1.44, \quad q(1) = 0.75, \quad \text{gap} = 0.69.$$

At $x = 1, \lambda = 2$, the gap is zero. No feasible point can have a smaller objective than the matching lower bound.

A certificate of optimality

Suppose an algorithm returns:

1. a feasible point with objective 10.003,
2. and a dual-feasible point with bound 10.000.

Then the unknown optimum lies in

$$10.000 \leq p^* \leq 10.003.$$

The solution is within 0.003 of optimal, even if we do not know p^* exactly.

For a minimization problem, a dual-feasible point gives...

A. an upper bound on p^*

B. a lower bound on p^*

C. always the exact optimizer

D. a learning rate

Correct: B — a lower bound on p^*

Why: Weak duality gives $q(\lambda) \leq p^*$ for every dual-feasible multiplier.

Four conditions at once

The general Lagrangian

For

$$\min_{\boldsymbol{x}} f(\boldsymbol{x})$$

subject to $g_i(\boldsymbol{x}) \leq 0, \quad h_j(\boldsymbol{x}) = 0,$

define

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \sum_i \lambda_i g_i(\boldsymbol{x}) + \sum_j \nu_j h_j(\boldsymbol{x}).$$

Inequality multipliers satisfy $\lambda_i \geq 0$. Equality multipliers ν_j may have either sign.

KKT 1 — primal feasibility

The candidate must obey the original problem:

$$g_i(\mathbf{x}^*) \leq 0, \quad h_j(\mathbf{x}^*) = 0.$$

No amount of stationarity can rescue an infeasible point.

For the running example, primal feasibility is simply

$$x^* - 1 \leq 0.$$

KKT 2 — dual feasibility

Each inequality multiplier must obey

$$\lambda_i^* \geq 0.$$

This sign makes the dual function a valid lower bound for a minimization problem.

There is no sign restriction on equality multipliers.

KKT 3 — stationarity

The Lagrangian must be stationary in the primal variables:

$$\nabla f(\mathbf{x}^*) + \sum_i \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

The objective gradient is balanced by normals from the active constraints.

This is L19's gradient-alignment equation with more possible normals.

KKT 4 — complementary slackness

For every inequality,

$$\lambda_i^* g_i(\mathbf{x}^*) = 0.$$

Two permitted cases:

state	$g_i(\mathbf{x}^*)$	λ_i^*
inactive	< 0	$= 0$
active	$= 0$	≥ 0

The four conditions on one slide

primal feasibility	$g_i(\mathbf{x}^*) \leq 0, \quad h_j(\mathbf{x}^*) = 0$
dual feasibility	$\lambda_i^* \geq 0$
stationarity	$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}$
complementary slackness	$\lambda_i^* g_i(\mathbf{x}^*) = 0$

Feasible here, feasible there, gradients balance, and inactive constraints cost zero.

Necessary or sufficient?

Under a suitable constraint qualification, a local optimum satisfies KKT: the conditions are **necessary**.

For a convex problem—convex f and g_i , affine h_j —any KKT point is globally optimal: the conditions are **sufficient**.

Under Slater's condition (a strictly feasible point exists), strong duality also holds:

$$d^* = p^*.$$

We use these results; a proof belongs in a convex optimization course.

KKT by hand

The problem and its Lagrangian

$$\min_x (x - 2)^2 \quad \text{subject to} \quad x - 1 \leq 0.$$

$$\mathcal{L}(x, \lambda) = (x - 2)^2 + \lambda(x - 1).$$

KKT requires

$$x - 1 \leq 0, \quad \lambda \geq 0,$$

$$2(x - 2) + \lambda = 0, \quad \lambda(x - 1) = 0.$$

Case 1 — assume the constraint is inactive

If $x - 1 < 0$, complementary slackness forces $\lambda = 0$.

Stationarity then gives

$$2(x - 2) = 0 \Rightarrow x = 2.$$

But $x = 2$ violates $x \leq 1$.

The inactive case is inconsistent with primal feasibility.

Case 2 — the constraint is active

Set $x - 1 = 0$, so $x = 1$.

Stationarity gives

$$2(1 - 2) + \lambda = 0 \Rightarrow \lambda = 2.$$

Dual feasibility holds because $\lambda = 2 \geq 0$.

$$x^* = 1, \quad \lambda^* = 2, \quad p^* = 1.$$

Verify all four receipts

condition	at $(x, \lambda) = (1, 2)$	check
primal feasibility	$1 - 1 = 0 \leq 0$	✓
dual feasibility	$2 \geq 0$	✓
stationarity	$2(1 - 2) + 2 = 0$	✓
complementary slackness	$2(1 - 1) = 0$	✓

The problem is convex, so this KKT point is the global optimum.

The multiplier again measures sensitivity

Replace the boundary $x \leq 1$ by $x \leq b$ near $b = 1$.

$$V(b) = (b - 2)^2 \quad \text{for } b < 2.$$

$$V'(b) = 2(b - 2), \quad V'(1) = -2.$$

Our multiplier is $\lambda^* = 2$. With $g(x; b) = x - b$,

$$\frac{dV}{db} = -\lambda^*.$$

Increasing the allowed boundary by 0.01 decreases the optimum by about 0.02.

Check the inactive example

$$\min_x (x - 0.2)^2 \quad \text{subject to} \quad x - 1 \leq 0.$$

The unconstrained minimizer $x^* = 0.2$ is feasible with slack 0.8.

Complementary slackness therefore gives $\lambda^* = 0$.

Stationarity checks:

$$2(0.2 - 0.2) + 0 = 0.$$

Moving the boundary slightly has no effect on the optimum.

L19's example, now with an inequality budget

L19 maximized xy on the line $x + y = 1$. Keep the budget but allow spending less of it:

$$\max_{x,y} xy \quad \text{subject to} \quad x + y \leq 1, \quad x \geq 0, \quad y \geq 0.$$

In this deck's standard form: minimize $f(x, y) = -xy$ with

$$g_1 = x + y - 1 \leq 0, \quad g_2 = -x \leq 0, \quad g_3 = -y \leq 0.$$

$$\mathcal{L} = -xy + \lambda(x + y - 1) - \mu_1 x - \mu_2 y, \quad \lambda, \mu_1, \mu_2 \geq 0.$$

L19 wrote $\mathcal{L} = xy - \lambda(x + y - 1)$ for the maximization. Minimizing $-xy$ with $+\lambda$ is the same bookkeeping; the numbers below agree.

Does the budget bind?

Stationarity:

$$-y + \lambda - \mu_1 = 0, \quad -x + \lambda - \mu_2 = 0.$$

Suppose $x > 0$ and $y > 0$. Complementary slackness gives $\mu_1 = \mu_2 = 0$, so

$$x = y = \lambda.$$

1. Budget inactive: $\lambda = 0$ forces $x = y = 0$, contradicting $x > 0$.

2. Budget active: $x + y = 1$ gives $x = y = \frac{1}{2}$ and $\lambda = \frac{1}{2}$.

The product is $xy = \frac{1}{4}$, exactly L19's answer — the inequality binds, so the equality solution carries over.

Verify KKT at $(1/2, 1/2)$

condition	at $(x, y) = (\frac{1}{2}, \frac{1}{2}), \lambda = \frac{1}{2}, \mu = 0$	check
primal feasibility	$\frac{1}{2} + \frac{1}{2} - 1 = 0 \leq 0, \quad -\frac{1}{2} \leq 0$	✓
dual feasibility	$\lambda = \frac{1}{2} \geq 0, \quad \mu_1 = \mu_2 = 0$	✓
stationarity	$(-\frac{1}{2} + \frac{1}{2}, -\frac{1}{2} + \frac{1}{2}) = (0, 0)$	✓
complementary slackness	$\lambda \cdot 0 = 0, \quad \mu_i x_i = 0$	✓

The budget's price is $\lambda = \frac{1}{2}$, the same shadow price L19 computed: raising the budget by δ raises the best product by about $\delta/2$.

$(0, 0)$ also satisfies KKT with every multiplier zero — $-xy$ is not convex, so KKT points are candidates. Compare values: $1/4$ beats 0 .

If $g_i(x^*) < 0$, what must complementary slackness imply?

A. $\lambda_i^* < 0$

B. $\lambda_i^* = 0$

C. $\lambda_i^* = 1$

D. $\nabla f = 0$ always

Correct: B — $\lambda_i^* = 0$

Why: The product $\lambda_i^* g_i(x^*)$ can vanish with nonzero slack only when its multiplier is zero.

A non-negative allocation problem

Project preferred amounts onto a budget

Three channels prefer amounts

$$\mathbf{a} = (3, 1, 0.5).$$

Choose a non-negative allocation of total $B = 2.5$ that stays closest to these preferences:

$$\min_{\mathbf{x}} \quad \frac{1}{2} \sum_{i=1}^3 (x_i - a_i)^2$$

$$\text{subject to} \quad \sum_i x_i = 2.5, \quad x_i \geq 0.$$

This is Euclidean projection onto a simplex-like set.

Write the inequalities in standard form

Use $-x_i \leq 0$ with multipliers $\mu_i \geq 0$, and multiplier ν for the equality:

$$\mathcal{L} = \frac{1}{2} \sum_i (x_i - a_i)^2 + \nu \left(\sum_i x_i - B \right) - \sum_i \mu_i x_i.$$

Stationarity in coordinate i gives

$$x_i - a_i + \nu - \mu_i = 0.$$

Positive coordinates

If $x_i > 0$, the inequality $-x_i \leq 0$ is inactive.

Complementary slackness gives $\mu_i = 0$, so

$$x_i = a_i - \nu.$$

Every positive coordinate is reduced by the same threshold ν .

Zero coordinates

If $x_i = 0$, stationarity gives

$$\mu_i = \nu - a_i.$$

Dual feasibility $\mu_i \geq 0$ requires $\nu \geq a_i$.

Combining the two cases:

$$x_i^* = \max(a_i - \nu, 0).$$

Determine the threshold

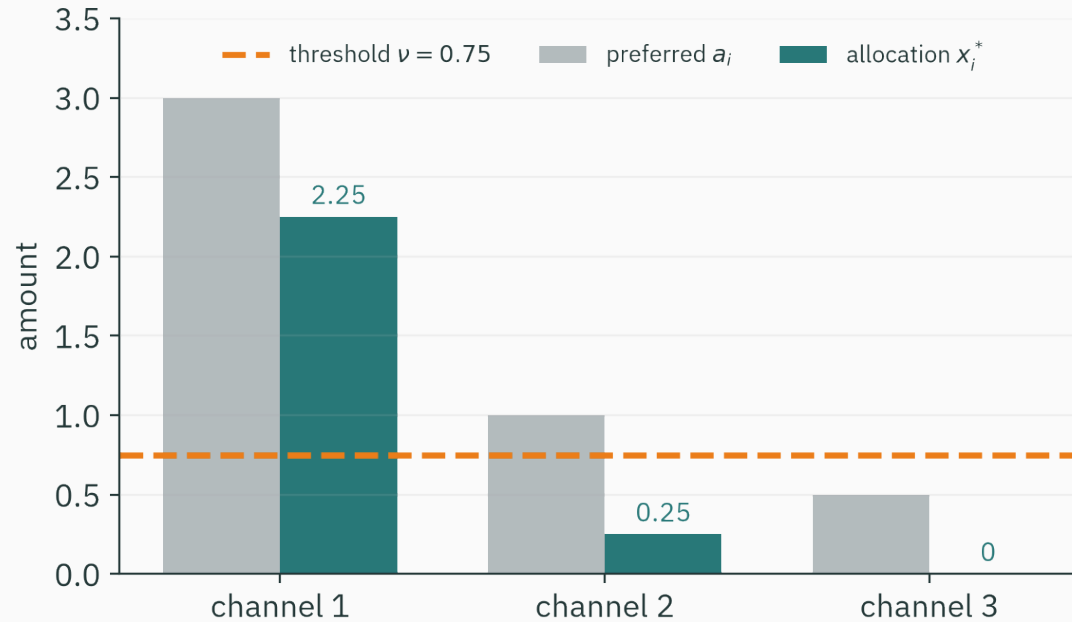
We still need $\sum_i x_i = 2.5$:

$$\max(3 - \nu, 0) + \max(1 - \nu, 0) + \max(0.5 - \nu, 0) = 2.5.$$

Try $0.5 < \nu < 1$. Then only the first two coordinates are positive:

$$(3 - \nu) + (1 - \nu) = 2.5.$$

$$\nu = 0.75.$$



$\mathbf{x}^* = (2.25, 0.25, 0)$ — the third coordinate is clipped at zero because its preferred amount lies below the common threshold.

Verify KKT numerically

For $\nu = 0.75$,

$$\mu = \max(\nu \mathbf{1} - \mathbf{a}, \mathbf{0}) = (0, 0, 0.25).$$

1. primal: $\mathbf{x} \geq 0$ and $\sum_i x_i = 2.5$,
2. dual: $\mu \geq 0$,
3. stationarity: $\mathbf{x} - \mathbf{a} + \nu \mathbf{1} - \mu = \mathbf{0}$,
4. complementary slackness: $\mu_i x_i = 0$ for each i .

The computation is a threshold search

```
import numpy as np

a = np.array([3.0, 1.0, 0.5])
B = 2.5
lo, hi = 0.0, a.max()

for _ in range(60):
    nu = (lo + hi)/2
    if np.maximum(a-nu, 0).sum() > B:
        lo = nu
    else:
        hi = nu

x = np.maximum(a-hi, 0)
print(x, x.sum())  # [2.25 0.25 0. ] 2.5
```

When the certificate is exact

Weak and strong duality

Weak duality always gives

$$d^* \leq p^*.$$

Strong duality means the best bound is exact:

$$d^* = p^*.$$

For convex problems satisfying a constraint qualification such as Slater's condition, strong duality holds.

Convexity turns KKT from a candidate test into a global optimality certificate.

Slater's condition, operationally

For a convex problem, look for a point satisfying

$g_i(\mathbf{x}) < 0$ for every inequality,

$h_j(\mathbf{x}) = 0$ for every equality.

Such a point is **strictly feasible**.

In the running problem $x \leq 1$, the point $x = 0$ is strictly feasible. Strong duality is therefore unsurprising.

Why solve a dual problem?

The dual can be useful because it may:

1. have fewer variables,
2. expose separable subproblems,
3. provide a numerical lower bound at every iteration,
4. give interpretable prices,
5. or reveal structure hidden in the primal.

Support vector machines are a standard ML example: data points receive dual weights, and only support vectors keep nonzero weights.

A practical stopping rule

Many solvers track three quantities:

1. primal residual: violation of original constraints,
2. dual residual: violation of stationarity,
3. duality gap: distance between primal value and dual bound.

Small residuals plus a small gap give a checkable stopping criterion.

Common sign errors

Before doing any algebra, record:

1. Is this a minimization or maximization problem?
2. Did we write inequalities as $g_i \leq 0$ or $g_i \geq 0$?
3. Which sign appears in the Lagrangian?
4. Which multipliers must be non-negative?

Do not memorize a multiplier sign separately from the chosen standard form.

The dual gives a bound; KKT explains when the bound and the feasible solution meet.

1. Weak duality: $d^* \leq p^*$ for minimization.
2. A zero duality gap certifies global optimality.
3. KKT combines primal feasibility, dual feasibility, stationarity, and complementary slackness.
4. An inactive inequality has zero multiplier; an active one may have a positive price.
5. For convex problems under regularity, KKT is sufficient and strong duality holds.

Practice

1. Solve $\min_x (x + 1)^2$ subject to $x \geq 0$ using KKT.
2. For $\min_x x^2$ subject to $x \leq 2$, state whether the constraint is active and find its multiplier.
3. For the running problem, evaluate $q(0)$, $q(1)$, $q(2)$, and $q(3)$.
4. Project $\mathbf{a} = (2, 1)$ onto $x_1 + x_2 = 1$, $x_i \geq 0$.

1. Write $-x \leq 0$. The solution is $x^* = 0$ with $\lambda^* = 2$.
2. $x^* = 0$ is strictly feasible; the constraint is inactive and $\lambda^* = 0$.
3. $q(\lambda) = \lambda - \frac{\lambda^2}{4}$, so the values are 0, 0.75, 1, 0.75.
4. $\nu = 1$, hence $x^* = \max(\mathbf{a} - \nu \mathbf{1}, \mathbf{0}) = (1, 0)$.

Next: recognizable problem classes

Lagrange multipliers and KKT are general conditions. Some objectives and constraints have enough structure for highly reliable solvers.

Next lecture:

1. linear objective + linear constraints $\rightarrow$ linear program,
2. quadratic objective + linear constraints $\rightarrow$ quadratic program.

Read before L21 — Boyd & Vandenberghe Ch 5 (guided: 5.1–5.5, pictures and boxed statements first); MML §7.2.

KKT is Lagrange with inequalities and receipts.

Next: **Linear & Quadratic Programming** — the two constrained shapes with industrial-strength solvers.